

JEE MAIN REVISION MASTER: DUAL NATURE OF RADIATION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 25 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Photoelectric Effect & Einstein's Equation	17 PYQs	Quantum Nature
High	de Broglie Wavelength of Matter Waves	8 PYQs	Matter Waves

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Photoelectric Effect & Einstein's Equation	$e V_0 = h\nu - \phi, \quad K_{\max} = \frac{hc}{\lambda} - \phi, \quad \text{slope of } V_0 \text{ vs } \nu = \frac{h}{e}$
de Broglie Wavelength of Matter Waves	$\lambda = \frac{h}{\sqrt{2mqV}}, \quad \lambda_{e} = \frac{1.227}{\sqrt{V}} \text{ nm}, \quad \lambda_p : \lambda_{\alpha} = \frac{\sqrt{m_{\alpha} q_{\alpha}}}{\sqrt{m_p q_p}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Photoelectric Effect & Einstein's Equation (Very High Yield — 17 PYQs)

Core Principle: Einstein's photoelectric equation: $K_{\max} = e V_0 = h\nu - \phi = h(\nu - \nu_0)$. Work function $\phi = h\nu_0 = (hc / \lambda_0)$. Photoelectric current is directly proportional to light intensity. Maximum kinetic energy $K_{\max}$ and stopping potential V_0 depend strictly on frequency ν , completely independent of light intensity.

■■ **Caution / JEE Pitfall:** Increasing light intensity increases the number of emitted photoelectrons (current), but does NOT change their maximum kinetic energy or stopping potential!

■ **Speed Trick:** The slope of the graph of stopping potential V_0 versus frequency ν is a universal constant (h / e), identical for all metals.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Variation of photoelectric current with anode potential is shown below. Choose the correct option ($V_0(0)$ = stopping potential)..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Work function of the metal is..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The anode voltage of a photocell is kept fixed. The frequency of the light falling on the cathode is gradually increased. Then the correct g..." [Ans: N/A]

3.2 de Broglie Wavelength of Matter Waves (High Yield — 8 PYQs)

Core Principle: de Broglie hypothesis: $\lambda = (h / p) = (h / mv) = \frac{h}{\sqrt{2mE}}$. For an electron accelerated through potential difference V : $\lambda = \frac{h}{\sqrt{2meV}} = \frac{1.227}{\sqrt{V}} \text{ nm} = \frac{12.27}{\sqrt{V}} \text{ \AA}$. For a gas molecule at temperature T : $\lambda = \frac{h}{\sqrt{3mkT}}$.

■■ **Caution / JEE Pitfall:** For proton (m, q) vs α -particle ($4m, 2q$) accelerated through same potential V : $(\lambda_p / \lambda_\alpha) = \sqrt{(m_\alpha q_p / m_p q_\alpha)} = \sqrt{(4 \times 2)} = \sqrt{8} = 2\sqrt{2}$.

■ **Speed Trick:** Remember the quick electron formula: Accelerated through 100 V, $\lambda = \frac{1.227}{\sqrt{100}} = 0.1227 \text{ nm} = 1.227 \text{ \AA}$.

Representative Source PYQs:

- **[2025 Shift] Q#3 (MCQ):** "If we consider an electron and a photon of same de-Broglie wavelength, then they will have same..." [Ans: N/A]
- **[2023 Shift] Q#3 (MCQ):** "A 60 W source emits monochromatic light of wavelength 662.5 nm. The number of photons emitted per second is..." [Ans: N/A]
- **[2022 Shift] Q#3 (MCQ):** "The de-Broglie wavelength of a particle of kinetic energy K is λ , the wavelength of the particle, if its kinetic energy $\frac{K}{4}$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Variation of photoelectric current with anode potential is shown below. Choose the correct option (V_0 = stopping potential)

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: In the photoelectric effect, photoelectric current changes with anode potential in this way: 1. At zero anode potential, the current is still present. This happens because the emitted photoelectrons already have some kinetic energy, so some of them can reach the anode even without extra attraction. 2. When the anode is given a negative potential, it repels the photoelectrons. So, fewer electrons reach the anode and the current decreases. 3. At a particular negative potential, even the fastest photoelectrons cannot reach the anode. The current becomes zero at this point. This potential is called

Benchmark Problem 2: KCET 2025 (Q#3)

If we consider an electron and a photon of same de-Broglie wavelength, then they will have same

- (A) Angular momentum
- (B) Energy
- (C) Velocity
- (D) Momentum

Correct Answer: None

Solution Analysis: They'll have the same momentum. Reasoning (from de Broglie's relation): $\lambda = (h / p)$ $\Rightarrow p = (h / \lambda)$. If an electron and a photon share the same λ , they share the same p .